CARDSTOCK ROCKET ENGINEERING & PRE-FLIGHT QUALITY CONTROL

PART 1: PRECISION FABRICATION, CENTER OF MASS / PRESSURE & SAFETY CERTIFICATION

Unit 2: Kinematics & Dynamics
Standards: HS-PS2-1 HS-ETS1-2

STUDENT NAME: _____ **PERIOD:** _____ **DATE:** _____ **ROCKET NAME / CALL SIGN:** _____

🛠️ **TWO-STAGE IN-CLASS FABRICATION SEQUENCE:** Tuesday: Fuselage & Fin Build | Wednesday (30-min Min Day): Assembly, Ballast & Pre-Flight QC

📐 STEP-BY-STEP FABRICATION PROTOCOL & PHYSICAL TOLERANCES

ENGINEERING DESIGN PROCESS

Step 1: Fuselage Rolling & Tolerance Fit [Tuesday Build]

Wrap cardstock tightly around the 1/2-inch PVC rolling mandrel. Secure longitudinal seam with 3 thin pieces of clear tape. Slide tube back and forth to verify smooth travel without friction binding.

> ✓ **Gravity Drop Test: Tube must slide freely off vertical mandrel under its own weight.**

Step 2: Fin Geometry & Symmetry [Tuesday Build]

Cut 3 or 4 identical cardstock fins using the precision template. Crease 5 mm mounting tabs. Align fins parallel to fuselage centerline spaced at 120° (3 fins) or 90° (4 fins).

> ✓ **Symmetry Check: Fins must be rigidly perpendicular (no twists that induce barrel roll).**

Step 3: Nose Cone & Clay Ballast [Wednesday QC]

Roll cardstock into a smooth aerodynamic nose cone. Pack a calibrated 5–10 g lump of modeling clay firmly into the tip. Secure nose cone to fuselage with an airtight tape wrap.

> ✓ **Airtight Seal Check: Gently blow into fuselage base; zero air should leak through nose cone.**

Step 4: String-Loop Balance Test [Wednesday QC]

Tie a slip-knot string loop around the rocket. Adjust loop until rocket balances horizontally. Mark this physical **Center of Mass (CM)**. Verify CM is 1–2 tube diameters forward of fins.

> ✓ **Swing Test: Spin balanced rocket overhead on 1m string; it must fly nose-forward, not tumble.**

⚖️ PRE-FLIGHT PHYSICAL SPECIFICATIONS & MASS BUDGET

RECORD WITH ELECTRONIC BALANCE & METRIC RULER

1. Total Rocket Mass (m):	2. Total Length (L):	3. CM Location:	4. Stability Margin:
_____ grams = _____ kg	Length: _____ cm	CM: _____ cm from nose	CM is _____ cm forward of fins
(Divide grams by 1,000 for kg)	Tip of nose to tail base	String balance point	Must be > 2.5 cm (1 diameter)

🛡️ PRE-FLIGHT QUALITY CONTROL CHECKLIST & FLIGHT CERTIFICATION

REQUIRED BEFORE THURSDAY OUTDOOR LAUNCH

- ☐ **Fuselage Sliding Fit:** Slides off 1/2" test mandrel freely under own weight (no friction binding).
- ☐ **Nose Cone Airtight Seal:** Sealed with tape; zero air leakage during base blow test.
- ☐ **Aerodynamic Fin Symmetry:** 3 or 4 rigid cardstock fins firmly attached parallel to tube axis.
- ☐ **Positive Stability Margin:** String test confirms Center of Mass (CM) is forward of Center of Pressure (CP).
- ☐ **Mass Budget Target:** Total dry mass in 20–30 g sweet spot (light enough to accelerate, heavy enough to coast).

PRE-FLIGHT CERTIFIED
Teacher Quality Inspection Stamp

_____
Instructor Signature & Launch Clearance

CARDSTOCK ROCKET FLIGHT TELEMETRY & KINEMATIC ANALYSIS

Student: _____ **Period:** _____

PART 2: MULTI-TIMER LAUNCH TELEMETRY, APOGEE DERIVATIONS & ENGINEERING SYNTHESIS

Launch Pad Telemetry Squad: • Pad Safety Officer | • Pressurization Tech (Calibrated PSI) | • Master Trigger | • 3 Stopwatch Timers | • Recovery Officer

⏱️ LAUNCH PAD FLIGHT TELEMETRY LOG (THURSDAY LAUNCH TRIALS)

OPERATING PRESSURE: CALIBRATED LAUNCHER PSI (40–80 PSI)

TRIAL #	TIMER 1 (S)	TIMER 2 (S)	TIMER 3 (S)	AVERAGE T_{HANG} (S)	STABILITY (1–5)	TRAJECTORY NOTES
Trial 1					★ ★ ★ ★ ★	Straight / Spiral / Tumble
Trial 2					★ ★ ★ ★ ★	Straight / Spiral / Tumble
Trial 3 (Opt)					★ ★ ★ ★ ★	Straight / Spiral / Tumble

** Note: Multi-timer averaging cancels out individual student reaction time error (±0.20 s). Select your **Best Official Trial** for calculations below.*

🧱 QUANTITATIVE FLIGHT KINEMATICS CALCULATIONS (FRIDAY ANALYSIS)

G = 10 M/S² DOWNWARD

1. Time to Apogee (t_{up}) `t_up = t_hang / 2`

Flight symmetry: ascent time equals descent time.

2. Launch Exit Velocity (v_0) `v₀ = 10 · t_up`

At apogee v = 0, so $0 = v_0 - gt_{up} \rightarrow v_0 = 10 \cdot t_{up}$ (m/s).

3. Maximum Apogee Altitude (h_{max}) `h_max = 5 · (t_up)²`

Free-fall displacement: $h = \frac{1}{2}gt^2 = 5(t_{up})^2$ (meters). Convert to ft (×3.28).

4. Launch Tube Acceleration (a_{tube}) `a_tube = v₀² / (2 · L)`

Tube length L = 0.40 m. Use $v_0^2 = 2aL \rightarrow a_{tube} = v_0^2 / 0.80$ (m/s²).

📝 CLAIM-EVIDENCE-REASONING (CER) FLIGHT PERFORMANCE SYNTHESIS

SUMMATIVE EVALUATION

Prompt: Did your rocket achieve high-performance vertical flight, and which engineering variable (mass budget, pneumatic seal tolerance, or fin stability margin) was most decisive in determining your apogee?

- **Claim:** Direct statement answering the prompt regarding your rocket's flight performance.
- **Evidence:** Cite quantitative telemetry data (Mass in g, Hang Time in s, Exit Velocity in m/s, Apogee in m).
- **Reasoning:** Explain the physics connecting your evidence to your claim using Newton's 2nd Law ($a = F_{net} / m$) or the Center of Mass vs. Center of Pressure stability rule.